

Decent Coaching Classes
VIII SB
Mind Your Math - II
Answer Key: Test - Chapters 4, 5, 6 and 7.1

Q.1 (A) Choose the correct alternative

- (1) (C) 2 : 1
- (2) (B) $a^3 - 3a^2b + 3ab^2 - b^3$
- (3) (A) 3, 5
- (4) (A) 4

Q.1 (B) Solve the following (any four)

- (1) $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- (2) $(p + q)^3 = p^3 + 3p^2q + 3pq^2 + q^3$
- (3) seg XR is called an altitude of $\triangle XYZ$ (on side YZ).
- (4) $x^2 + 7x + 12 = (x + 3)(x + 4)$
- (5) $x = ky$

Q.2 Activities (any two)

- (1) $(2x + 3y + 4)^2 = 4x^2 + 9y^2 + 16 + 12xy + 24y + 16x$
[blanks: $4x^2$, 16, $24y$, $16x \rightarrow \text{sum} = 4x^2 + 9y^2 + 16 + 12xy + 24y + 16x$]
- (2) $5x^2 + 16x + 3 = 5x^2 + 15x + x + 3 = 5x(x + 3) + 1(x + 3) = (x + 3)(5x + 1)$
[blanks: 15, 1, $(x + 3)$, $(x + 3)$, $(x + 3)$]
- (3) $x \propto y \Rightarrow x = ky$. Using $x = 4$, $y = 20 \Rightarrow k = 1/5 \Rightarrow y = 5x$
Table: $x = 4, 6, 8, 10$ $y = 20, 30, 40, 50$ (blanks: $y = 30$ when $x = 6$; $x = 8$ when $y = 40$; $y = 50$ when $x = 10$)

Q.3 Solve the following (any four)

- (1) $(2p - 5)(2p + 7) = 4p^2 + 4p - 35$
- (2) $I(AG) : I(GD) = 2 : 1 \Rightarrow I(GD) = AG \div 2 = 10 \div 2 = 5 \text{ cm}$
- (3) $3x^2 - x - 4 = 3x^2 - 4x + 3x - 4 = x(3x - 4) + 1(3x - 4) = (3x - 4)(x + 1)$
- (4) Cost per pen = $225 \div 15 = ₹15$. Cost of 36 pens = $36 \times 15 = ₹540$
- (5) $(x + 3)^3 + (x - 3)^3 = 2x^3 + 6x(3)^2 = 2x^3 + 54x$
- (6) $8a^3 - 125b^3 = (2a)^3 - (5b)^3 = (2a - 5b)(4a^2 + 10ab + 25b^2)$

Q.4 Solve the following (any two)

- (1) Centroid property: G divides each median in ratio 2:1 from the vertex.
 - (i) $AD = 12 \Rightarrow AG = 8 \text{ cm}$, $GD = 4 \text{ cm}$
 - (ii) $BG = 8 \Rightarrow GE = 4 \text{ cm}$, $BE = BG + GE = 12 \text{ cm}$
 - (iii) The centroid divides each median in the ratio 2 : 1, measured from the vertex.
- (2) (i) $2x^2 - 7x - 15 = 2x^2 - 10x + 3x - 15 = 2x(x - 5) + 3(x - 5) = (x - 5)(2x + 3)$
 - (ii) $27p^3 + 1 = (3p)^3 + 1^3 = (3p + 1)(9p^2 - 3p + 1)$
 - (iii) $a^2 + 6a + 8 = (a + 2)(a + 4)$; $a^2 + 9a + 20 = (a + 4)(a + 5)$; $a^2 + 2a - 8 = (a + 4)(a - 2)$; $a^2 + 3a - 10 = (a + 5)(a - 2)$
Expression = $[(a + 2)(a + 4)/(a + 4)(a + 5)] \times [(a + 5)(a - 2)/(a + 4)(a - 2)] = (a + 2)/(a + 4)$
- (3) $k = 1500 \div 6 = 250$. Equation: $x = 250y$ (x = remuneration, y = number of labourers)
 - (ii) Remuneration for 25 labourers = $250 \times 25 = ₹6250$
 - (iii) Number of labourers = $4250 \div 250 = 17$
- (4) (i) $(3x + 2y + 5)^2 = 9x^2 + 4y^2 + 25 + 12xy + 20y + 30x$

$$(ii) (2m + 3n)^3 + (2m - 3n)^3 = 2(2m)^3 + 6(2m)(3n)^2 = 16m^3 + 108mn^2$$